



OrBit User Guide

Module 01: Getting Started & System Setup

PUBLIC DOCUMENTATION — FOR DEVELOPERS USERS

1 Introduction to OrBit

OrBit is a next-generation, decentralized local-first collaboration engine designed for software engineering teams. Unlike cloud-centric code hosting platforms, OrBit operates directly on your local machine, utilizing direct Peer-to-Peer (P2P) mesh networking and hidden Shadow Git repositories to deliver instantaneous, real-time code synchronization without forcing raw code blobs onto third-party servers.

Key Architecture Concept

OrBit splits collaboration into two distinct planes:

- **Local Data Plane:** All code diffs, snapshot calculations, and file transfers execute locally using Rust and direct P2P mesh networking.
- **Central Control Plane:** A lightweight Go server manages identity verification, friend requests, presence heartbeats, and encrypted dead-drop relays for offline peers.

2 System Requirements

Before installing OrBit, verify that your workstation satisfies the following requirements:

3 Installation Guide

3.1 Step 1: Installing the OrBit Desktop Client

1. Download the latest release binary for your operating system:

- **Windows:** OrBit-Setup-0.1.0.exe or .msi

Component	Requirement
Operating System	Windows 10/11 (64-bit), macOS 12+ (Apple Silicon/Intel), Linux (Ubuntu 22.04+)
Memory	4 GB RAM minimum (8 GB recommended for large repositories)
Disk Space	250 MB for OrBit Desktop App & Local Daemon
Networking	Local Area Network (LAN) or Internet connection; Port 65337 available locally
IDE	Visual Studio Code (version 1.80.0 or higher)

Table 1: OrBit Client System Requirements

- **macOS:** OrBit-0.1.0.dmg
- **Linux:** OrBit-0.1.0.AppImage

2. Launch the installer. On Windows, OrBit installs cleanly as a GUI application without spawning background terminal windows. 3. Upon completion, OrBit will automatically launch its background daemon and place an icon in your system tray.

3.2 Step 2: Installing the VS Code Extension

To unlock deep IDE presence and real-time teammate cursors: 1. Open Visual Studio Code. 2. Navigate to the Extensions view (Ctrl+Shift+X / Cmd+Shift+X). 3. Select **Install from VSIX...** and select `apps/vscode-extension/orbit-vscode-0.1.0.vsix`. 4. Alternatively, execute from terminal:

```
1 code --install-extension orbit-vscode-0.1.0.vsix
```

4 Account Authentication & Licensing

OrBit uses a streamlined license-key authentication architecture.

1. Launch the OrBit Desktop Client.
2. On the **Authentication Page**, enter your assigned OrBit License Key.
3. Click **Authenticate Key**.
4. The client verifies your license against the control server (`https://orbit-server-7bou.onrender.com`) and receives a secure JWT token stored in your system's secure local key storage.

Zero Passwords

You do not need to remember passwords or configure SSH keys. Your identity is cryptographically anchored by your local Ed25519 keypair generated in `%APPDATA%/OrBit/daemon/identity.key`.

5 Registering Your First Workspace

Once authenticated, register a software repository with OrBit:

1. On the OrBit Dashboard, click **+ Create / Import Project**.

2. Browse and select your project's root folder (e.g., `C:\Users\admin\Desktop\my-app`).
3. OrBit will initialize a hidden bare repository inside `.orbit/shadow.git` and trigger an initial snapshot.
4. Your workspace is now live and ready for local-first peer synchronization!